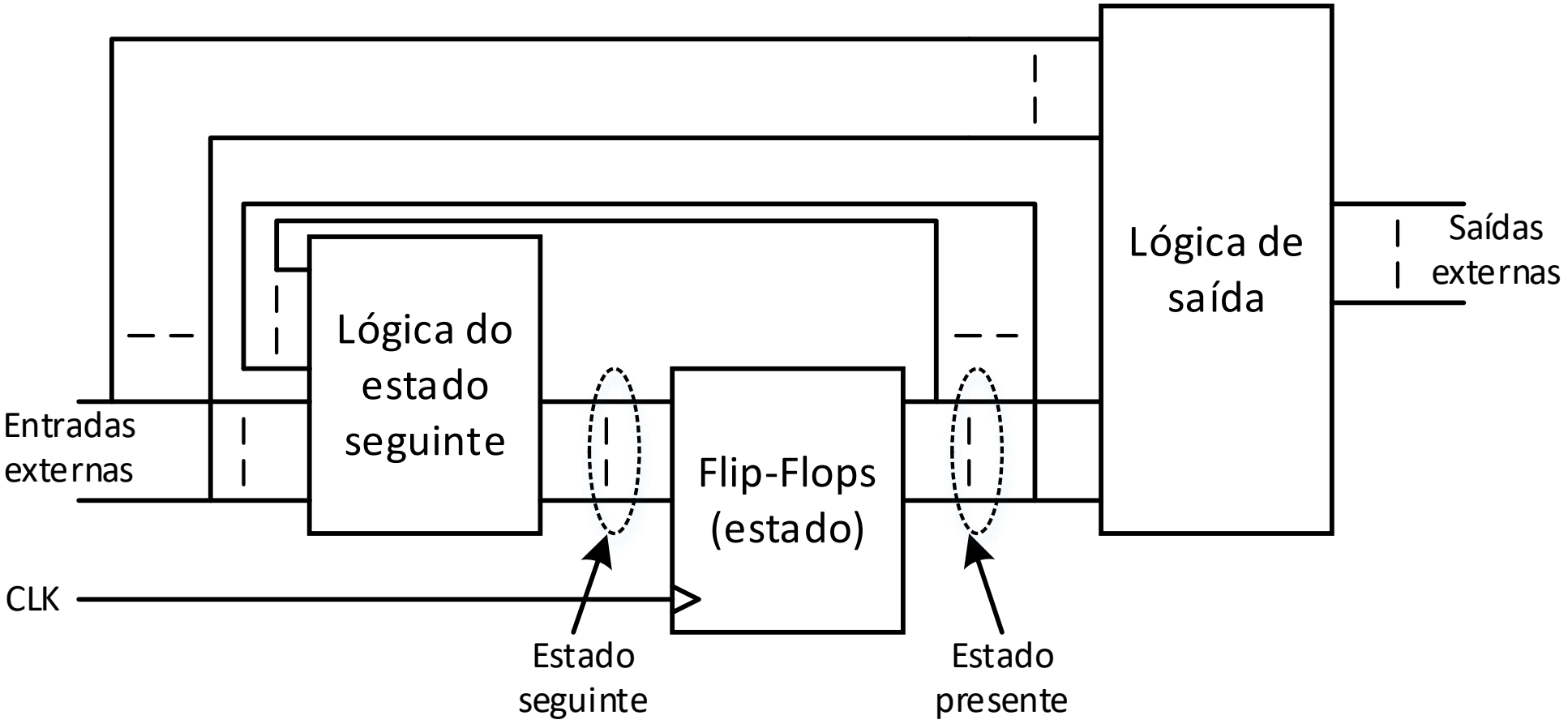
Circuitos Sequenciais: Análise Exercícios

Modelo de um Circuito Sequencial Síncrono



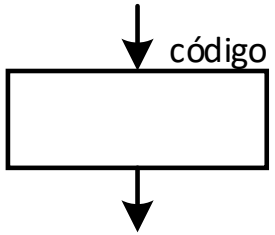
Análise de Circuitos Sequenciais Síncronos

Determinar a operação do circuito a partir do seu diagrama lógico:

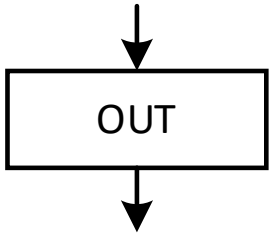
- Determina-se as equações da lógica de estado seguinte e da lógica de saída;
- Determinar a tabela de transição de estados e de saídas, em função do estado atual e das entradas externas;
- Obter tabela de estados codificados ou
- Em alternativa, obter o diagrama de estados – ASM Chart.

ASM (*Algorithmic State Machine*) *Chart* - Fluxograma

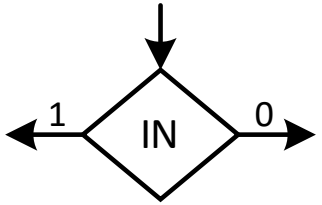
Estado



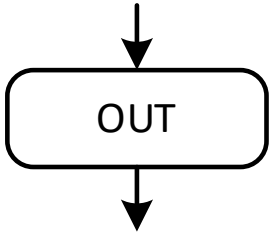
Saída Incondicional



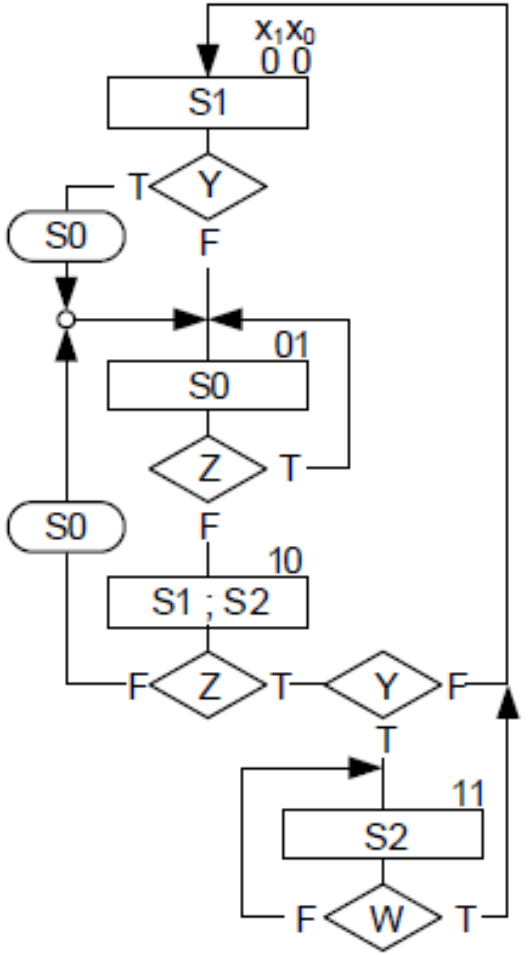
Entrada



Saída Condicional

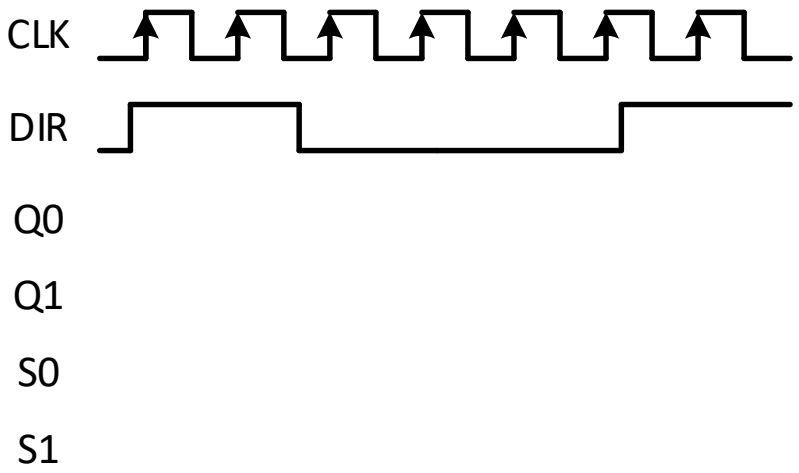


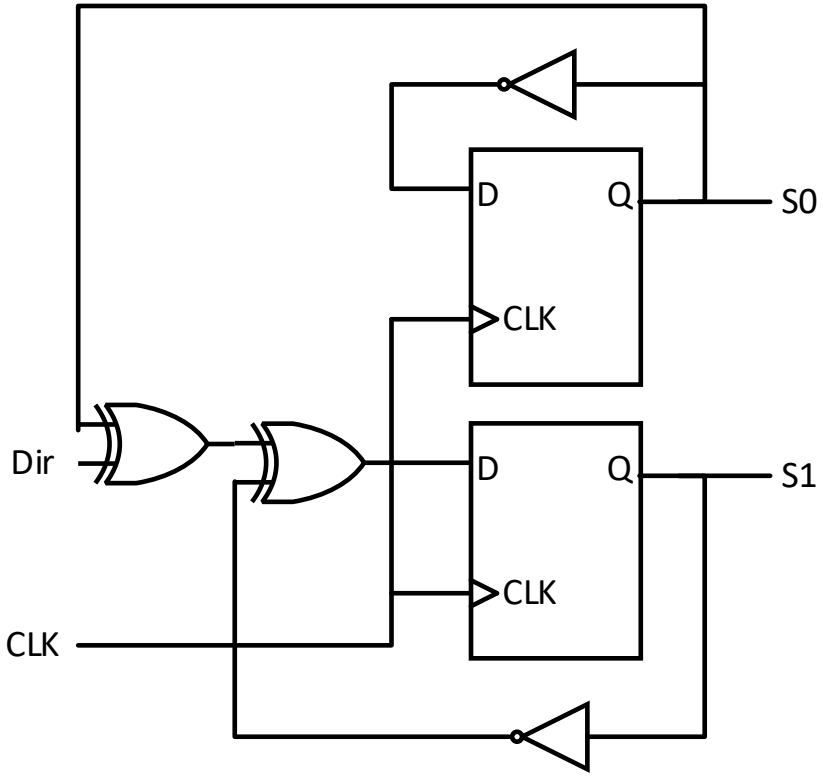
Exemplo



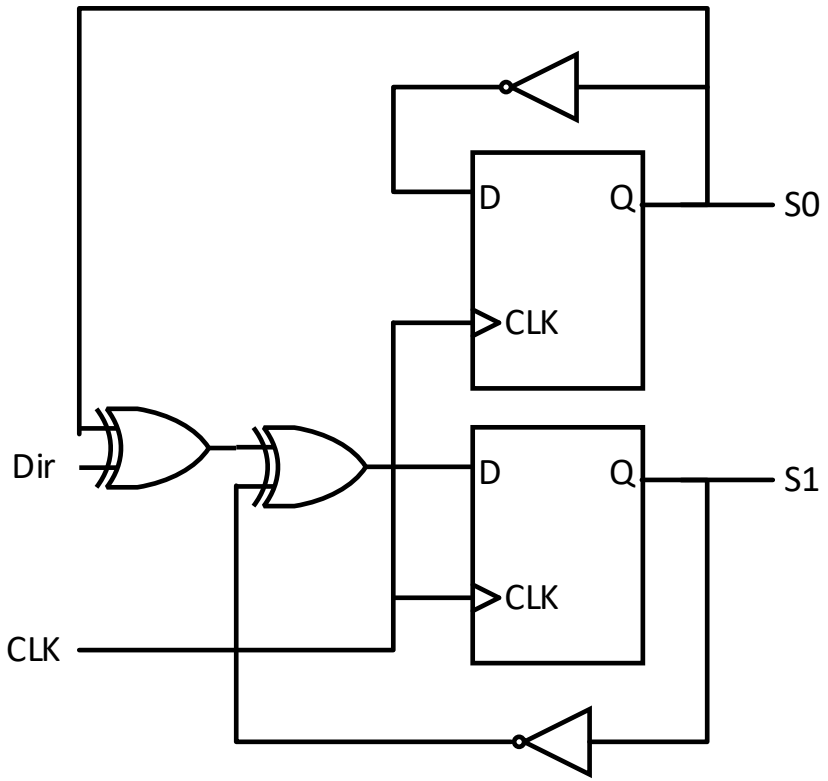
Exercício 1

- Determine tabela de transição de estados;
- Determine o fluxograma da máquina de estados;
- Desenhe o diagrama temporal.





Exercício 1 – Tabela de transição de estados



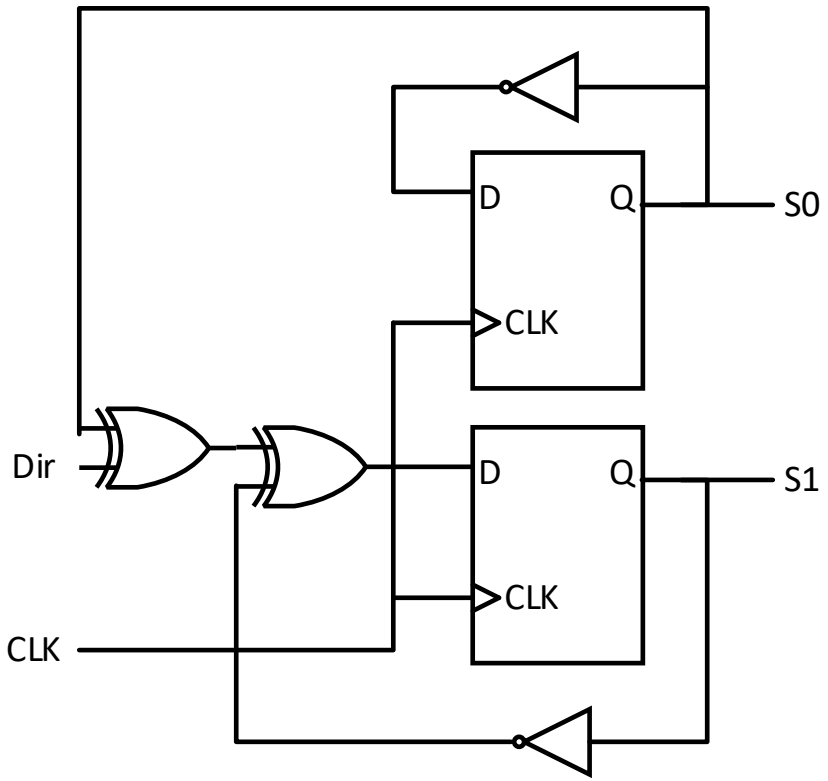
$D0 = \overline{Q0}$

$D1 = Dir \oplus Q0 \oplus \overline{Q1}$

$S0 = Q0$

$S1 = Q1$

Exercício 1 – Tabela de transição de estados



$D0 = \overline{Q0}$

$D1 = Dir \oplus Q0 \oplus \overline{Q1}$

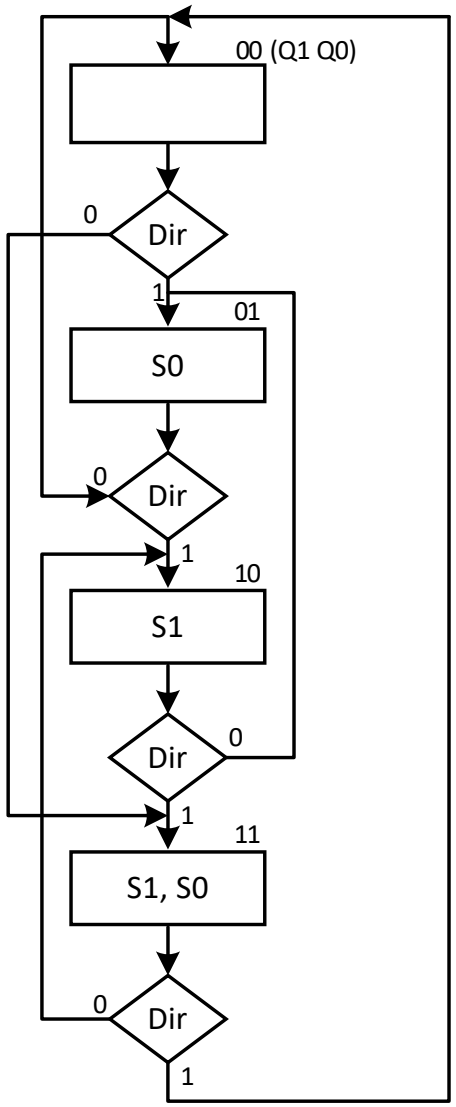
$S0 = Q0$

$S1 = Q1$

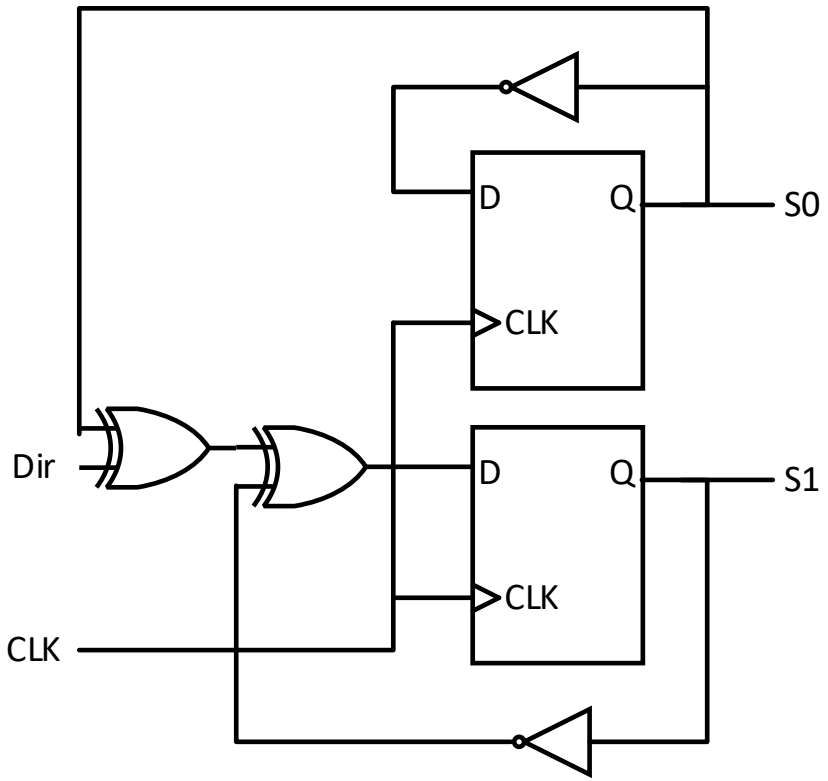
EP		Dir	ES		S0	S1
Q1	**Q0**	**-**	**D1**	**D0**	**-**	**-**
0	0	0	1	1	0	0
0	0	1	0	1	0	0
0	1	0	0	0	1	0
0	1	1	1	0	1	0
1	0	0	0	1	0	1
1	0	1	1	1	0	1
1	1	0	1	0	1	1
1	1	1	0	0	1	1

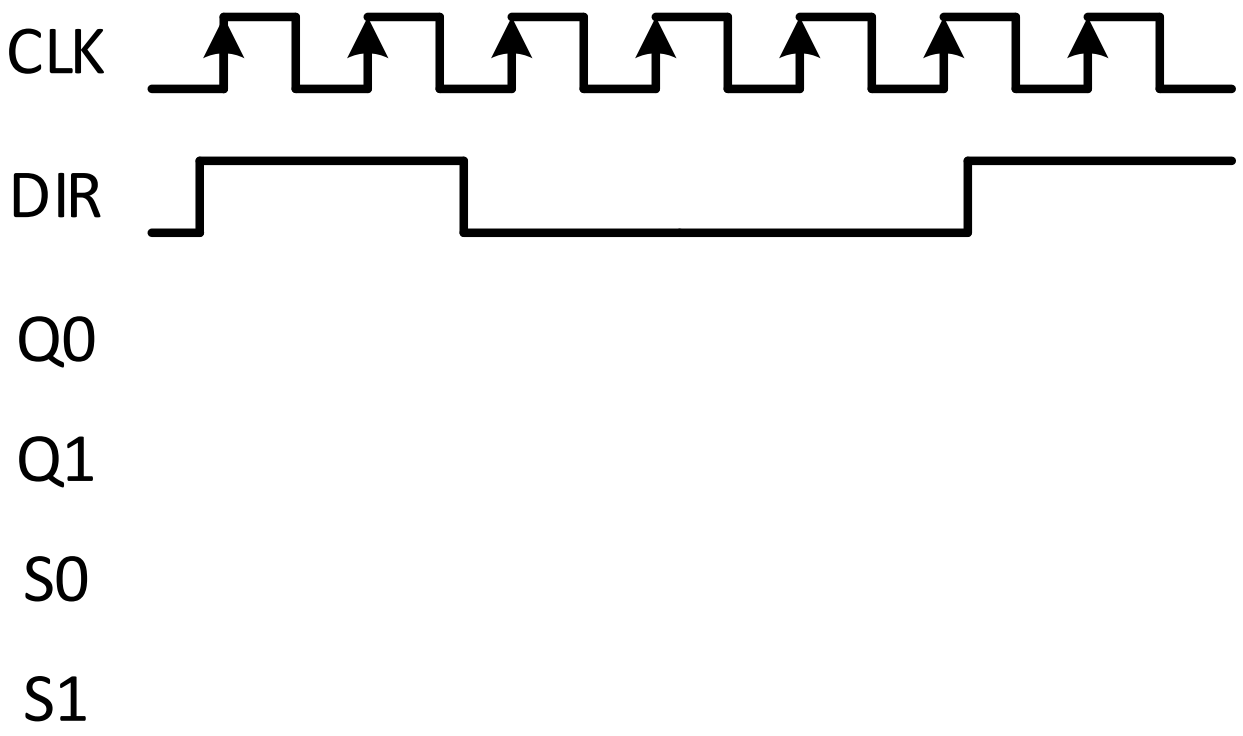
Exercício 1 – Fluxograma

EP		Dir	ES		S0	S1
Q1	Q0	-	D1	D0	-	-
0	0	0	1	1	0	0
0	0	1	0	1	0	0
0	1	0	0	0	1	0
0	1	1	1	0	1	0
1	0	0	0	1	0	1
1	0	1	1	1	0	1
1	1	0	1	0	1	1
1	1	1	0	0	1	1

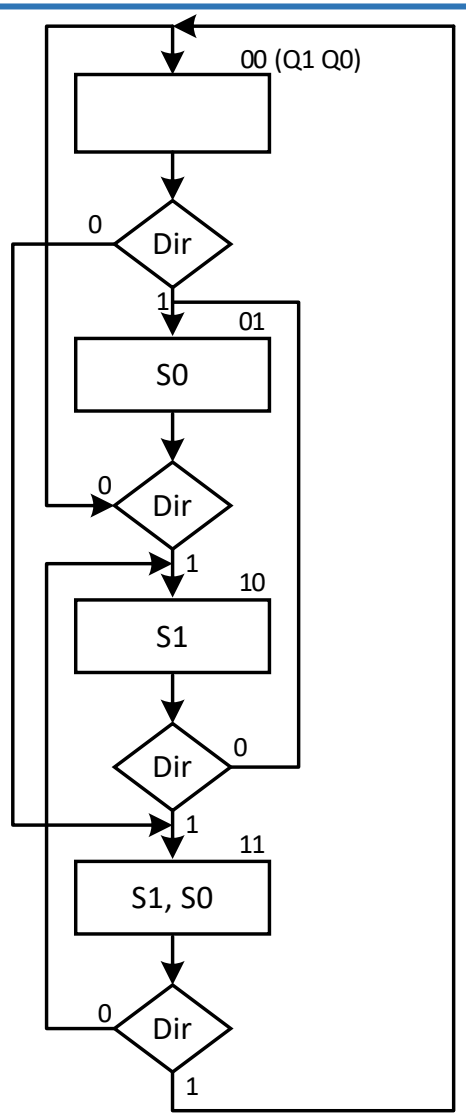


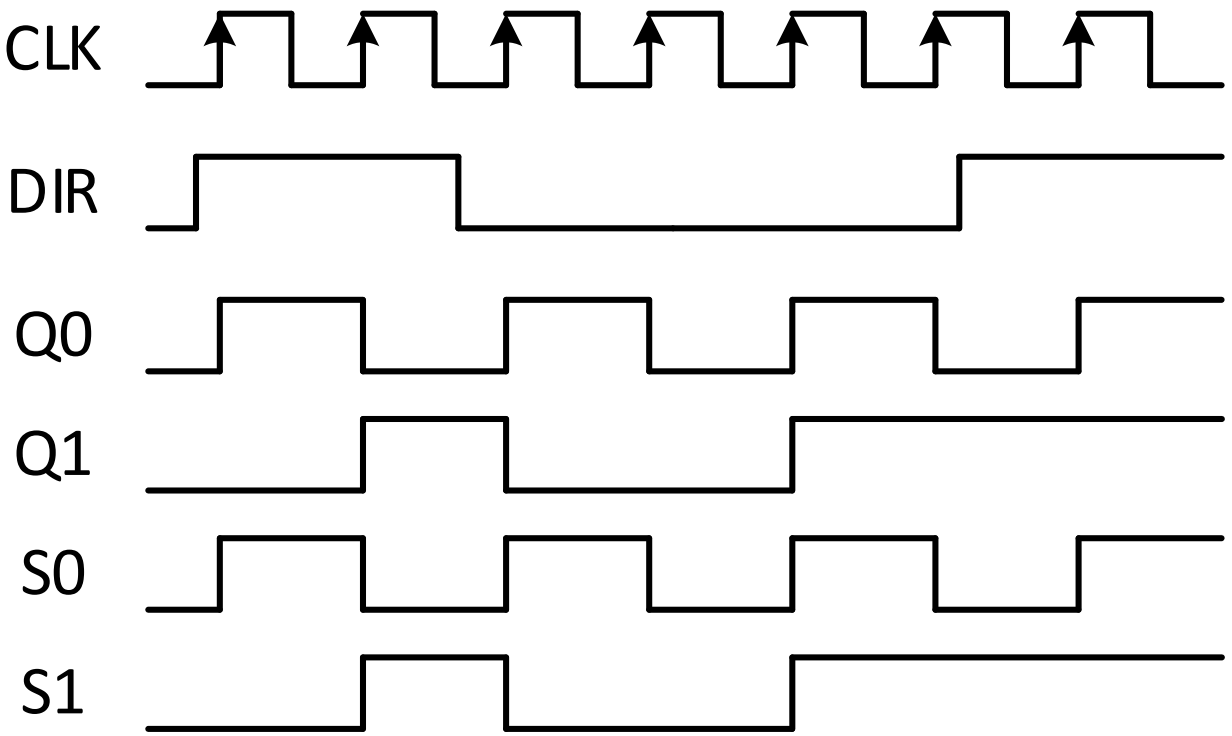
Exercício 1 – Diagrama Temporal





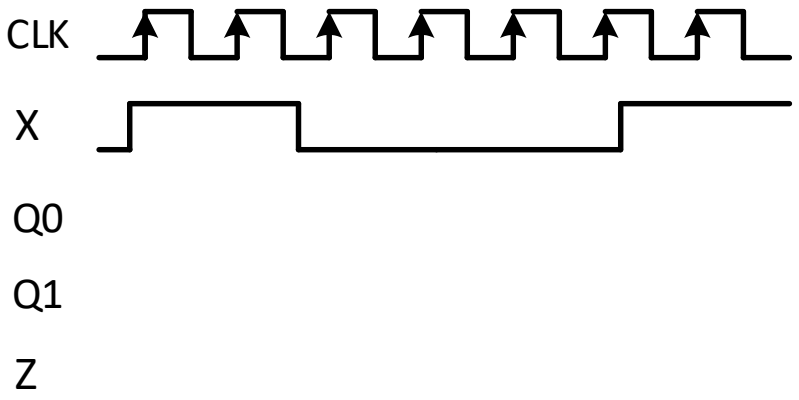
Exercício 1 – Diagrama Temporal

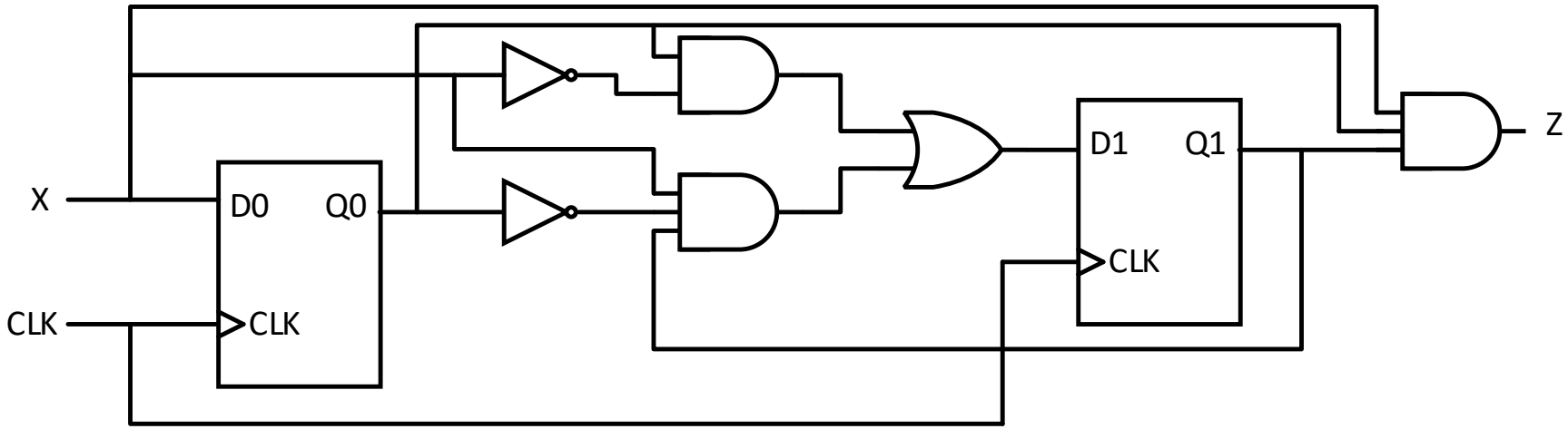




Exercício 2

- Determine tabela de transição de estados;
- Determine o fluxograma da máquina de estados;
- Desenhe o diagrama temporal.





Circuitos Sequenciais: Síntese

Síntese do Circuito Sequencial Síncrono

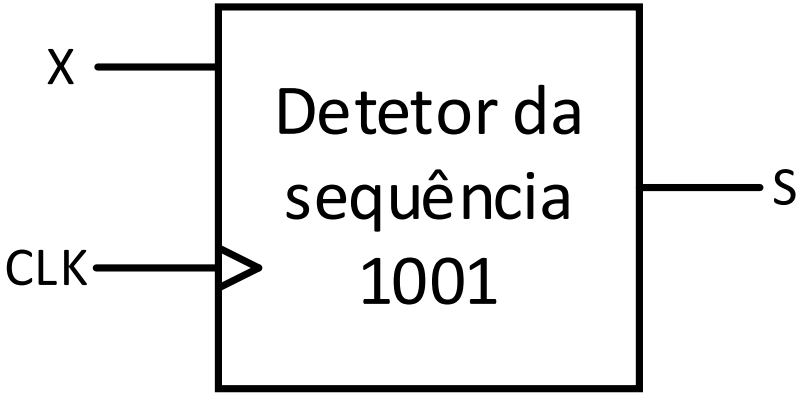
Passos para a síntese de um circuito sequencial síncrono:

- Determinar Fluxograma
- Diagrama de estados do circuito;
- Tabela de estados e de saída;
- Codificação dos estados;
- Tabela de excitações do circuito;
- Equações de estado seguinte e de saída;
- Logigrama ou esquema lógico.

Concepção de uma Máquina de Estados

Consideremos o exemplo seguinte:

- Pretende-se obter um circuito que identifique a ocorrência da sequência binária 1001 na sua entrada;
- Quando detetada, a saída deve vir a 1;
- O circuito lê um bit de entrada por cada ciclo de relógio.



Detetor de Sequência

00100100100101100010010010010011101 01

- Uma entrada (X)
- Uma saída (Z)
- Quatro estados (S0, S1, S2, S3)

Detetor de Sequência – Diagrama de Estados

00100100100101100010010010010011101 01

- Uma entrada (X)
- Uma saída (Z)
- Quatro estados (S0, S1, S2, S3)

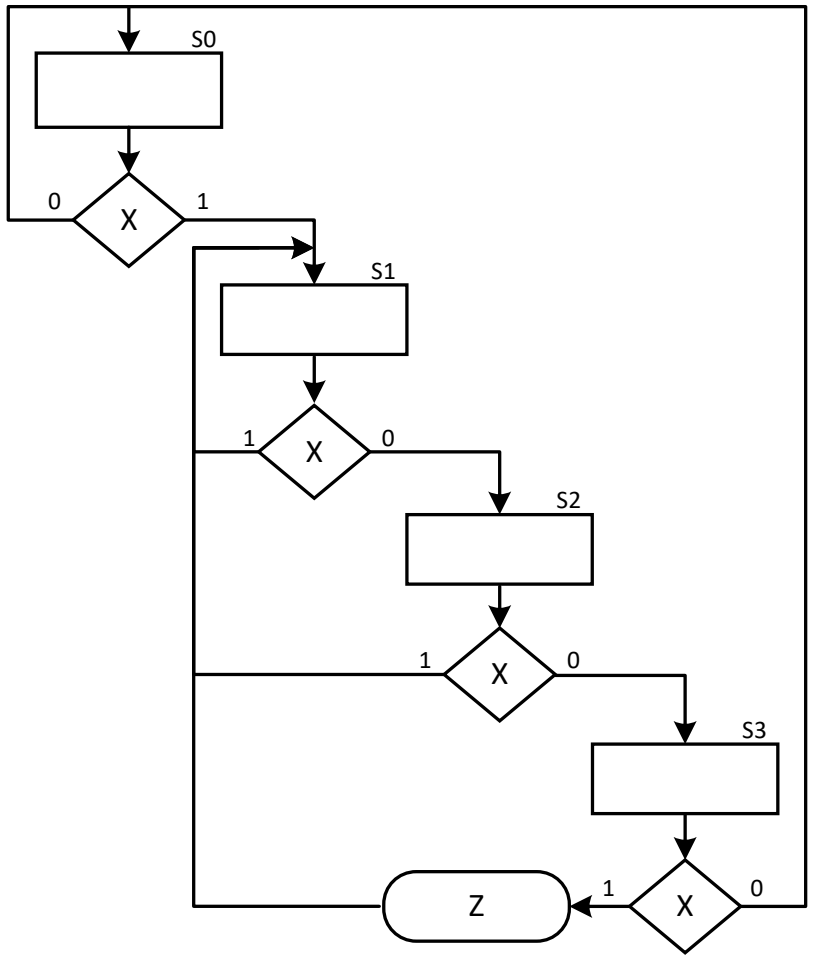
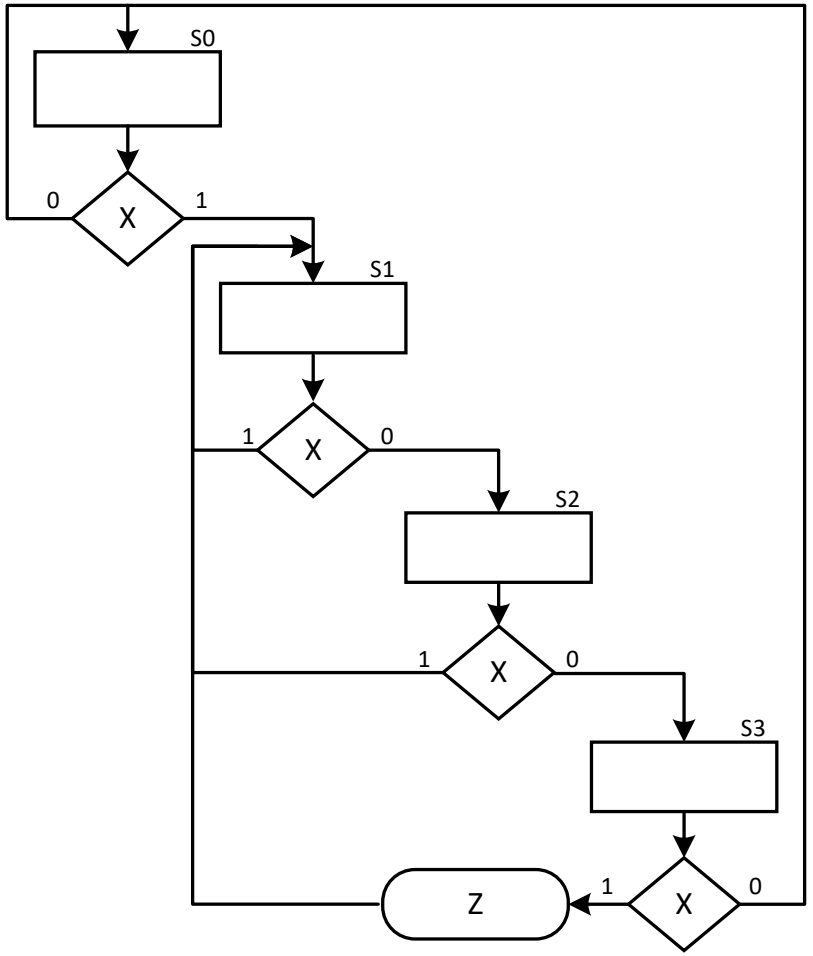
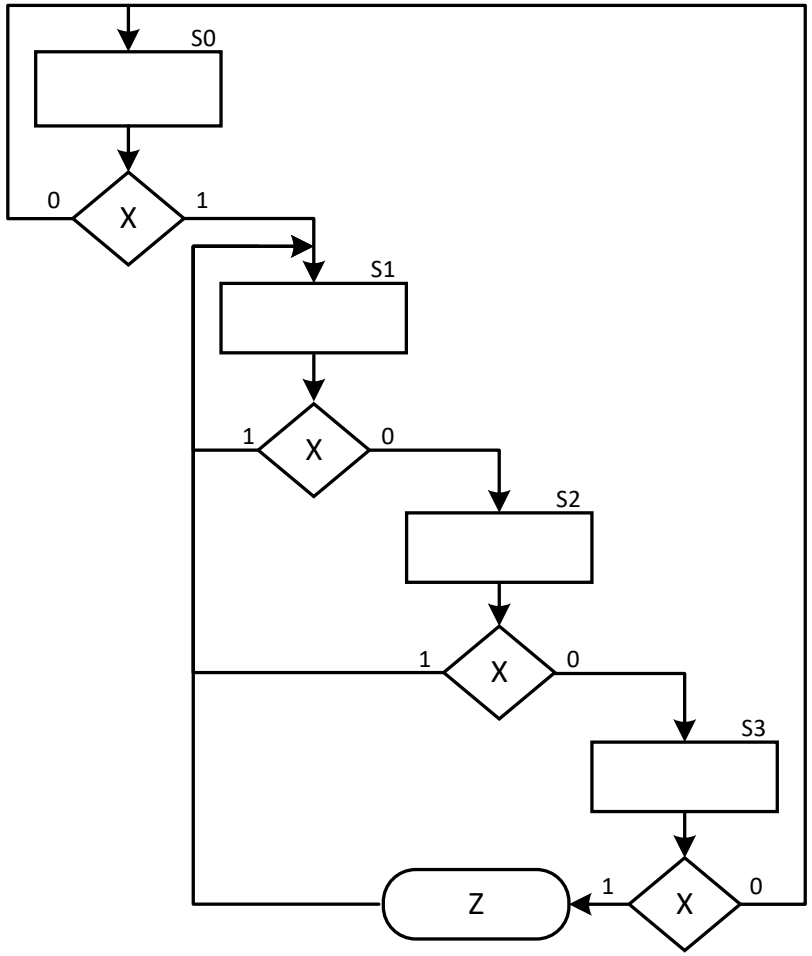


Tabela de Transição de Estados e de Saídas



EP	X	ES	Z
S0	0		
S0	1		
S1	0		
S1	1		
S2	0		
S2	1		
S3	0		
S3	1		

Tabela de Transição de Estados e de Saídas



EP	X	ES	Z
S0	0	S0	0
S0	1	S1	0
S1	0	S2	0
S1	1	S1	0
S2	0	S3	0
S2	1	S1	0
S3	0	S0	0
S3	1	S1	1

Codificação dos Estados

Estado		Nome
0	0	S0
0	1	S1
1	0	S2
1	1	S3

EP	X	ES	Z
S0	0	S0	0
S0	1	S1	0
S1	0	S2	0
S1	1	S1	0
S2	0	S3	0
S2	1	S1	0
S3	0	S0	0
S3	1	S1	1

EP		X	ES		Z

Codificação dos Estados

Estado		Nome
0	0	S0
0	1	S1
1	0	S2
1	1	S3

EP	X	ES	Z
S0	0	S0	0
S0	1	S1	0
S1	0	S2	0
S1	1	S1	0
S2	0	S3	0
S2	1	S1	0
S3	0	S0	0
S3	1	S1	1

EP		X	ES		Z
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	0	1	0
1	1	0	0	0	0
1	1	1	0	1	1

Escolha do FF; Tabela de Excitação dos FF

Vamos considerar uma implementação com FF tipo D

D	CLK	Q	QN
0	↑	0	1
1	↑	1	0

EP		X	ES		Z
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	0	1	0
1	1	0	0	0	0
1	1	1	0	1	1

EP		X	ES		Z
Q1	Q0	-	D1	D0	

Escolha do FF; Tabela de Excitação dos FF

Vamos considerar uma implementação com FF tipo D

D	CLK	Q	QN
0	↑	0	1
1	↑	1	0

EP		X	ES		Z
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	0	1	0
1	1	0	0	0	0
1	1	1	0	1	1

EP		X	ES		Z
Q1	Q0	-	D1	D0	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	0	1	0
1	1	0	0	0	0
1	1	1	0	1	1

Equações de Estado Seguinte e de Saída

EP		X	ES		Z
Q1	Q0	-	D1	D0	
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	0	1	0
1	1	0	0	0	0
1	1	1	0	1	1

Q1 \ Q0	$\overline{Q0}\,\overline{X}$	$\overline{Q0}\,X$	$Q0\,X$	$Q0\,\overline{X}$
$\overline{Q1}$	0	1	1	0
$Q1$	1	1	1	0

$$D0 = X + Q1\,\overline{Q0}$$

Q1 \ Q0	$\overline{Q0}\,\overline{X}$	$\overline{Q0}\,X$	$Q0\,X$	$Q0\,\overline{X}$
$\overline{Q1}$	0	0	0	1
$Q1$	1	0	0	0

$$D1 = Q1\,\overline{Q0}\,\overline{X} + \overline{Q1}\,Q0\,\overline{X}$$

$$Z = Q1\,Q0\,X$$

Logigrama da Máquina de Estados

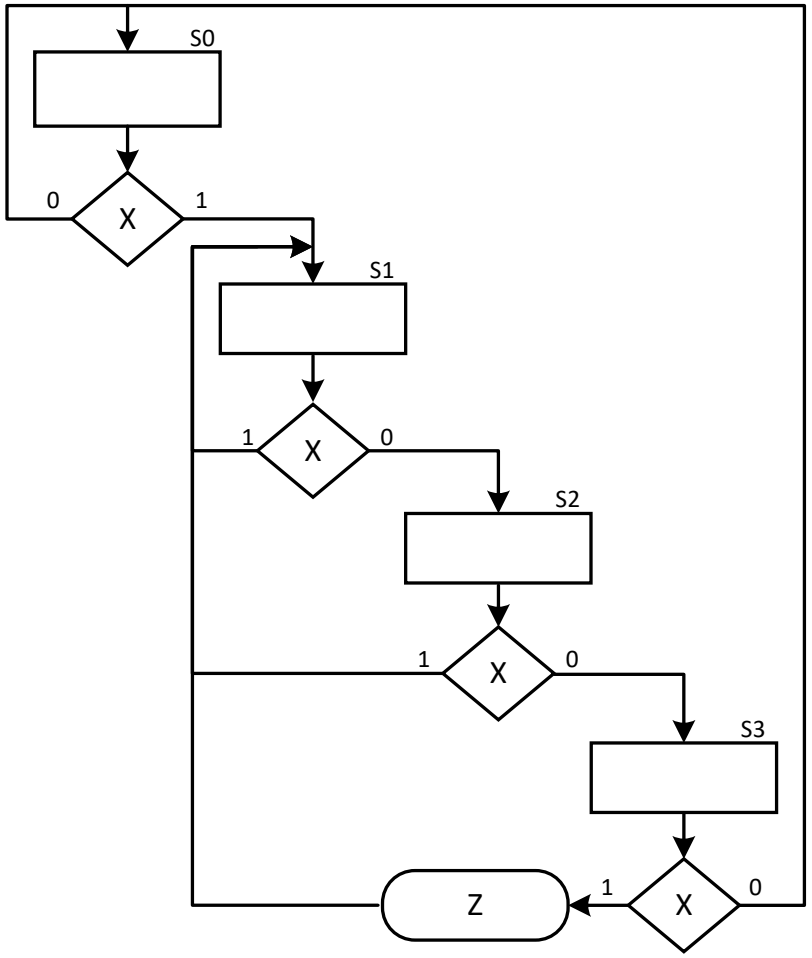
$D1 = Q1\,\overline{Q0}\,\overline{X} + \overline{Q1}\,Q0\,\overline{X}$

$D0 = X + Q1\,\overline{Q0}$

$Z = Q1\,Q0\,X$

?

Equações de Estado Seguinte e de Saída - Direto



Exercício – Implementar Máquina de Estados

4 estados
Duas entradas: Y, Z
Duas saídas: S0, S1

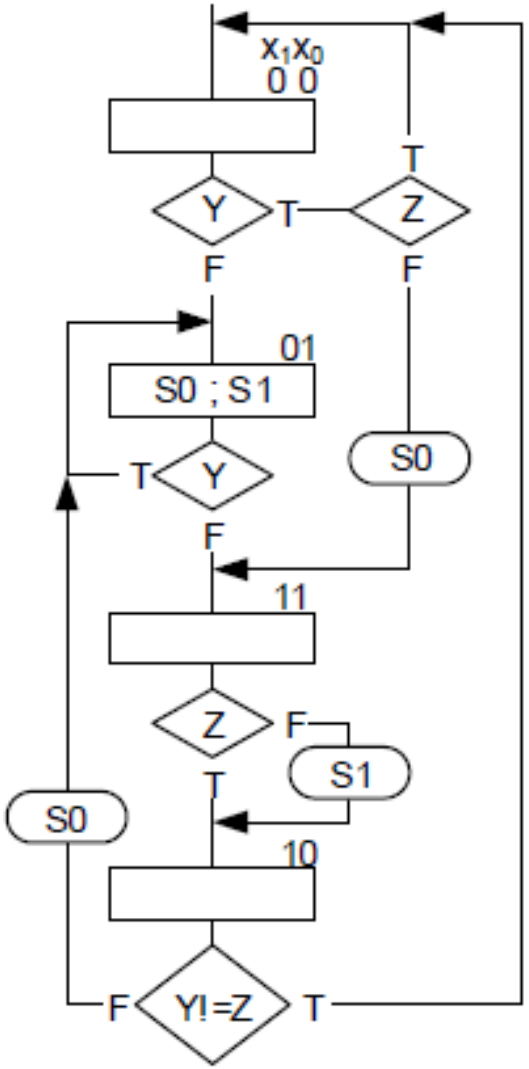
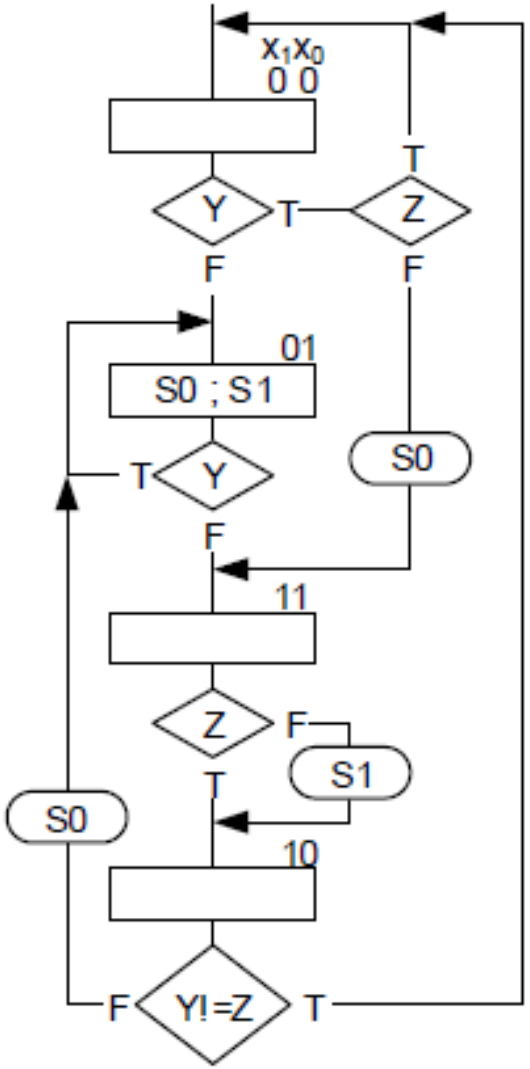
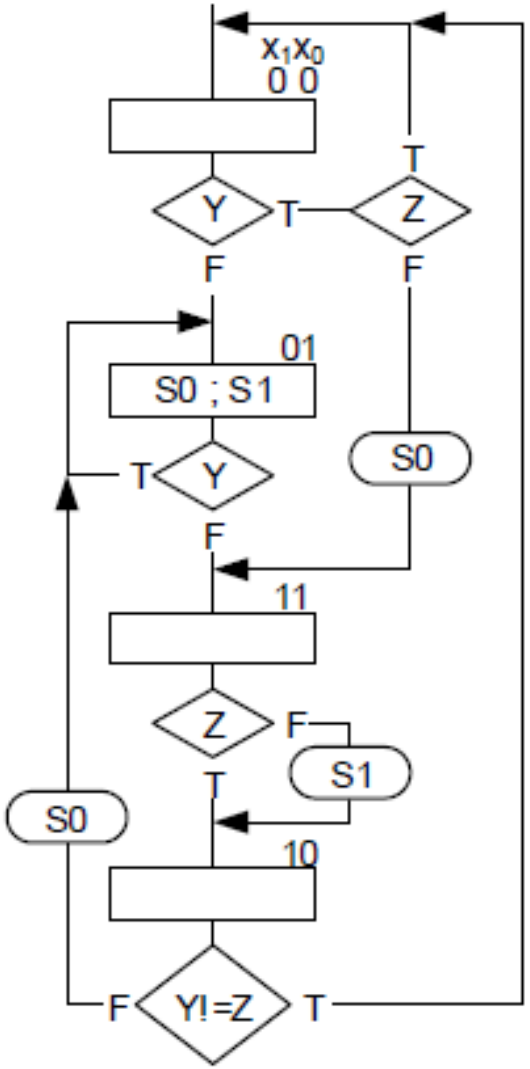


Tabela de Transição de Estados e de Saídas



EP X1 X0	Y	Z	ES	S0 S1
0 0	0	0		
0 0	0	1		
0 0	1	0		
0 0	1	1		
0 1	0	0		
0 1	0	1		
0 1	1	0		
0 1	1	1		
1 0	0	0		
1 0	0	1		
1 0	1	0		
1 0	1	1		
1 1	0	0		
1 1	0	1		
1 1	1	0		
1 1	1	1		

Tabela de Transição de Estados e de Saídas



EP X1 X0	Y	Z	ES		S0	S1
0 0	0	0	0	1	0	0
0 0	0	1	0	1	0	0
0 0	1	0	1	1	1	0
0 0	1	1	0	0	0	0
0 1	0	0	1	1	1	1
0 1	0	1	1	1	1	1
0 1	1	0	0	1	1	1
0 1	1	1	0	1	1	1
1 0	0	0	0	1	1	0
1 0	0	1	0	0	0	0
1 0	1	0	0	0	0	0
1 0	1	1	0	1	1	0
1 1	0	0	1	0	0	1
1 1	0	1	1	0	0	0
1 1	1	0	1	0	0	1
1 1	1	1	1	0	0	0

Tabela de Excitações

EP X1	X0	Y	Z	ES		S0	S1
0	0	0	0	0	1	0	0
0	0	0	1	0	1	0	0
0	0	1	0	1	1	1	0
0	0	1	1	0	0	0	0
0	1	0	0	1	1	1	1
0	1	0	1	1	1	1	1
0	1	1	0	0	1	1	1
0	1	1	1	0	1	1	1
1	0	0	0	0	1	1	0
1	0	0	1	0	0	0	0
1	0	1	0	0	0	0	0
1	0	1	1	0	1	1	0
1	1	0	0	1	0	0	1
1	1	0	1	1	0	0	0
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

EP X1	X0	Y	Z	ES D1	D0	S0	S1
0	0	0	0	0	1	0	0
0	0	0	1	0	1	0	0
0	0	1	0	1	1	1	0
0	0	1	1	0	0	0	0
0	1	0	0	1	1	1	1
0	1	0	1	1	1	1	1
0	1	1	0	0	1	1	1
0	1	1	1	0	1	1	1
1	0	0	0	0	1	1	0
1	0	0	1	0	0	0	0
1	0	1	0	0	0	0	0
1	0	1	1	0	1	1	0
1	1	0	0	1	0	0	1
1	1	0	1	1	0	0	0
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

Funções e Simplificação

EP X1	EP X0	Y	Z	ES D1	ES D0	S0	S1
0	0	0	0	0	1	0	0
0	0	0	1	0	1	0	0
0	0	1	0	1	1	1	0
0	0	1	1	0	0	0	0
0	1	0	0	1	1	1	1
0	1	0	1	1	1	1	1
0	1	1	0	0	1	1	1
0	1	1	1	0	1	1	1
1	0	0	0	0	1	1	0
1	0	0	1	0	0	0	0
1	0	1	0	0	0	0	0
1	0	1	1	0	1	1	0
1	1	0	0	1	0	0	1
1	1	0	1	1	0	0	0
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

Mapas de Karnaugh

EP X1 X0	Y Z	ES D1 D0	S0 S1
0 0	0 0	0 1	0 0
0 0	0 1	0 1	0 0
0 0	1 0	1 1	1 0
0 0	1 1	0 0	0 0
0 1	0 0	1 1	1 1
0 1	0 1	1 1	1 1
0 1	1 0	0 1	1 1
0 1	1 1	0 1	1 1
1 0	0 0	0 1	1 0
1 0	0 1	0 0	0 0
1 0	1 0	0 0	0 0
1 0	1 1	0 1	1 0
1 1	0 0	1 0	0 1
1 1	0 1	1 0	0 0
1 1	1 0	1 0	0 1
1 1	1 1	1 0	0 0

D1

X1 \ X0 \ Y Z	00	01	11	10
00	0	0	0	1
01	1	1	0	0
11	1	1	1	1
10	0	0	0	0

D0

X1 \ X0 \ Y Z	00	01	11	10
00	1	1	0	1
01	1	1	1	1
11	1	0	1	0
10	0	0	0	0

S1

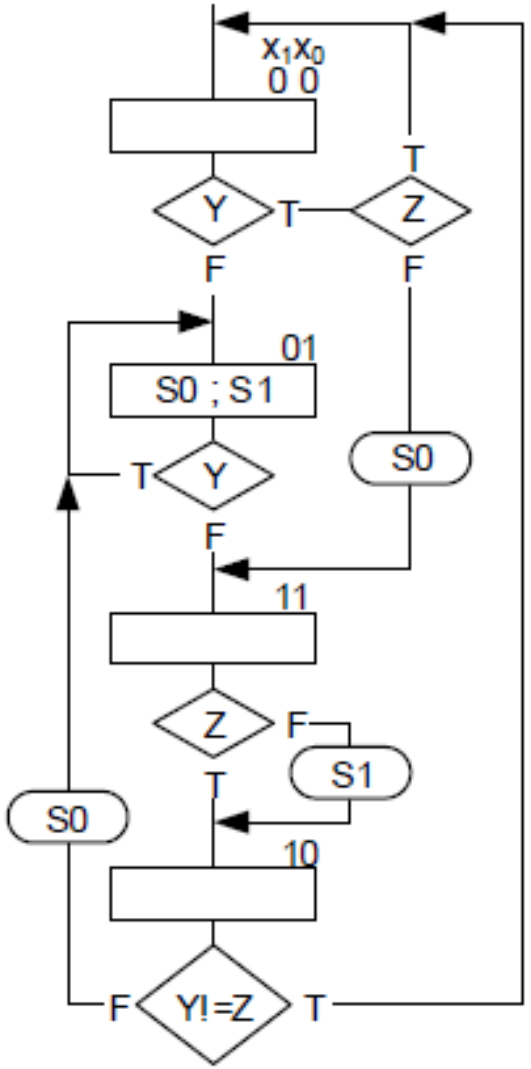
S0

X1 \ X0 \ Y Z	00	01	11	10
00	0	0	0	1
01	1	1	1	1
11	0	0	0	0
10	1	0	1	0

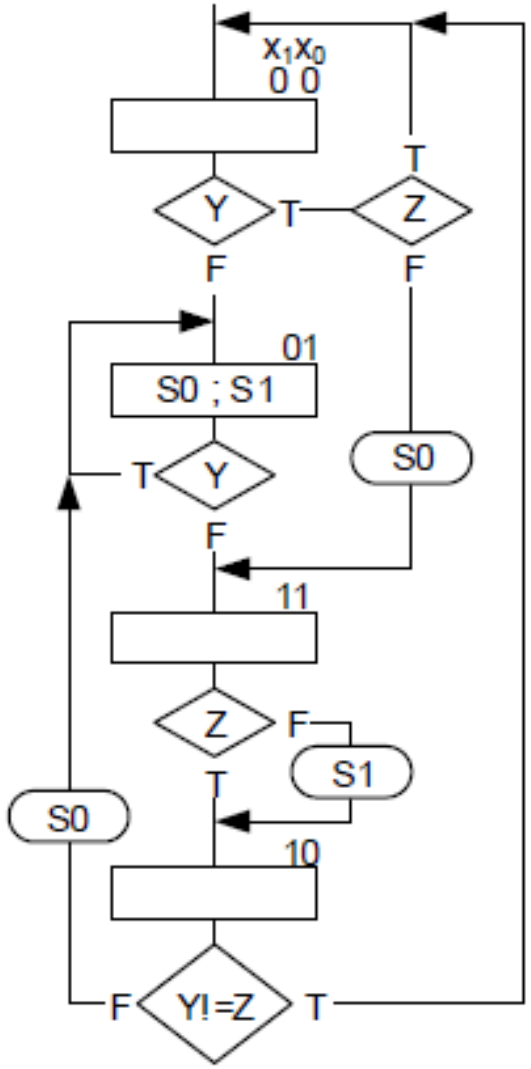
S1

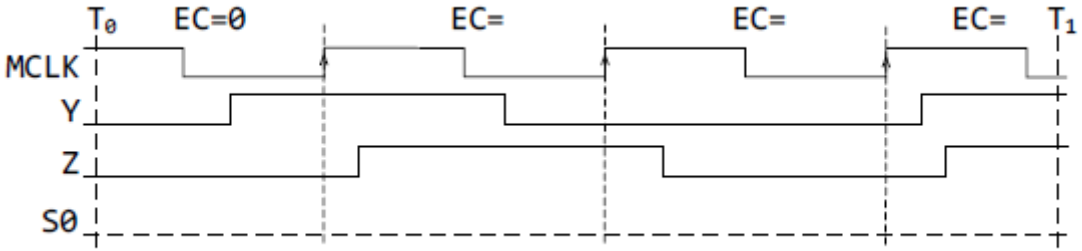
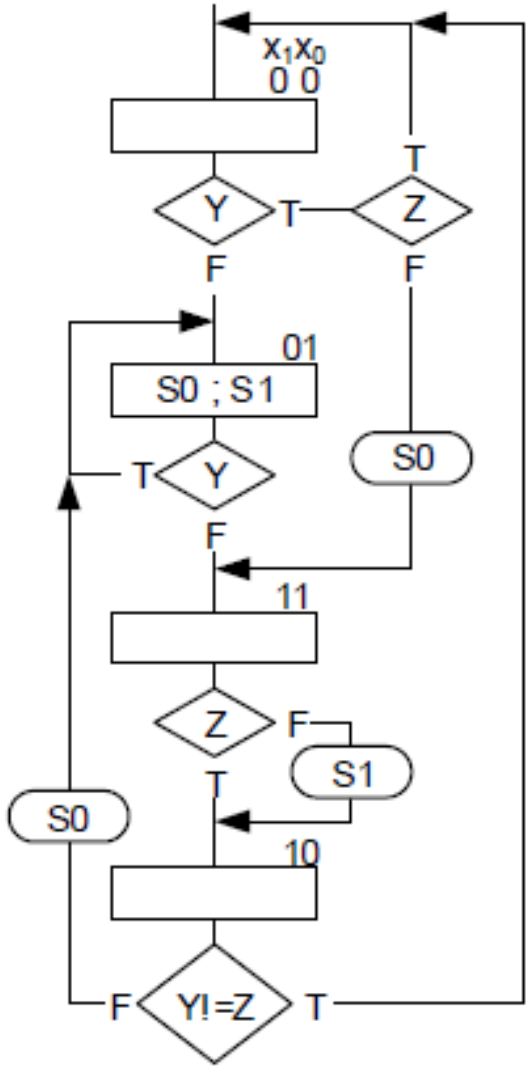
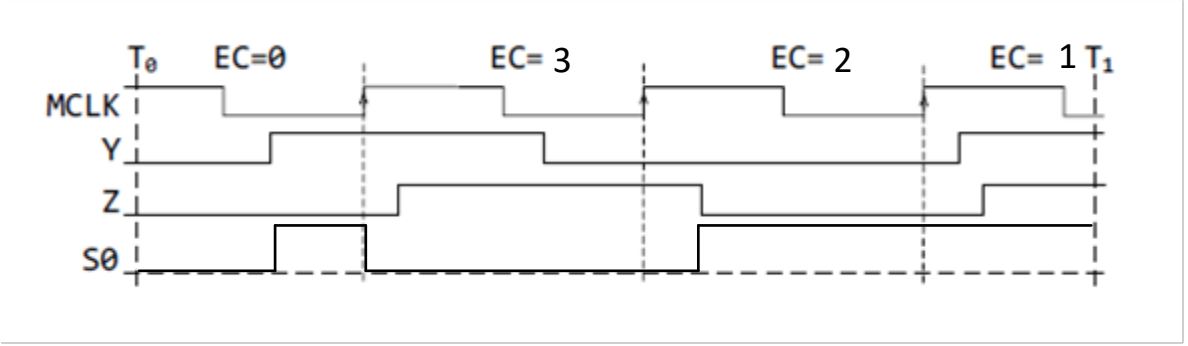
X1 \ X0 \ Y Z	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	1	0	0	1
10	0	0	0	0

Equações de Estado Seguinte e de Saída - Direto



Exercício – Análise Temporal





Exercício – Análise Temporal